

PROJECT REPORT: PARAKH

Part 1 : Project Overview

Project Name: Parakh (MetrologyLens AI)

Parakh is an AI-powered compliance and inspection system designed to enforce the Legal Metrology (Packaged Commodities) Rules, 2011. It allows government inspectors and auditors to scan packaged commodities using a mobile application. The system leverages OCR and advanced Large Language Models (LLMs) to extract text from product labels and automatically validate if the required mandatory declarations (like MRP, Net Quantity, Manufacturer Details, Consumer Care, and Manufacturing Date) are present and compliant.

Part 2 : System Architecture & Tech Stack

1. Mobile Application (Frontend):

- Framework: Flutter (Dart)
- Key Features:
 - * Multi-image Capture & Gallery Upload.
 - * Bilingual Support (English & Hindi) powered by a custom SettingsService.
 - * Dynamic Theming (Light/Dark mode) that seamlessly adapts to the system theme.
 - * Dynamic Dashboard showing Daily Targets, Completed Inspections, and Pending tasks.
 - * Real-time Local History tracking with image and compliance violation logging.
 - * Beautiful UI with responsive Cards and color-coded compliance status (Green for Pass, Red for Fail).

2. API Gateway & AI Engine (Backend):

- Framework: Python, FastAPI
- Key Features:
 - * Endpoints for image upload, OCR processing, and Rule Validation.
 - * Integrates with Google Gemini (or similar LLMs) for highly accurate layout extraction.
 - * Evaluates missing parameters against Legal Metrology Rules (e.g., Rule 6).
 - * Generates beautiful 'Parakh Inspection PDF' reports for formal notices.

3. Web Dashboard:

- Framework: HTML, CSS, JavaScript (served via FastAPI)
- Purpose: Acts as a Central Monitoring Portal for higher authorities to track live compliance scores.

Part 3 : Core Workflows

A. The Inspection Flow:

1. The Field Inspector logs into the Mobile App.
2. Navigates to the 'Scan' tab and clicks photos of a product's label.
3. Clicks 'Analyze'. The images are sent to the FastAPI Backend via Multipart Upload.
4. The Backend extracts details (Net Qty, MRP, Phone, Email, Date) and checks for violations.
5. The App instantly displays a '100% COMPLIANT' or 'NON-COMPLIANCE DETECTED' card.
6. The scan is saved locally in the 'History' tab and updates the 'Completed Today' dashboard metric.

B. Formal Legal Actions:

- If a product fails, the Inspector can click 'Download PDF'.
- The backend serves a heavily detailed official 'Show Cause Notice' / 'Inspection Report'.

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- This PDF can be shared or printed for penal actions under the Legal Metrology Act, 2009.

Part 4 : Recent Technical Enhancements

- UI/UX Polishing: Solved dark mode invisible text issues by switching from hardcoded Hex values to Flutter's dynamic `Theme.of(context).colorScheme`.
- Layout Fixes: Resolved UI overflow and spacing issues in the Scan Screen.
- History Management: Built an entirely new `HistoryService` to store complex objects (Inspection Results + Images) locally and display them in a dedicated list view.
- Dynamic Stats: Connected the Home Screen dashboard to actual inspection history to give a real-world feel to the Inspector's 'Pending Tasks'.
- Branding: Integrated the official 'logo.jpg' into the top AppBar for a professional look.